

MÖSSBAUER STUDY ON THE THERMAL DECOMPOSITION OF POTASSIUM TRIS (OXALATO) FERRATE(III) TRIHYDRATE AND BIS (OXALATO) FERRATE(II) DIHYDRATE

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The thermal decompositions of $K_3Fe(ox)_3 \cdot 3 H_2O$ and $K_2Fe(ox)_2 \cdot 2 H_2O$ in nitrogen have been studied using Mössbauer spectroscopy, X-ray diffraction and thermal analysis methods in order to determine the nature of the solid residues obtained after each stage of decomposition. Particularly, after dehydration at $113^\circ C$, the ferric complex is reduced into a ferrous compound, with a quadrupole splitting of 3.89 mm/s, which corresponds to the anhydrous form of $K_2Fe(ox)_2 \cdot 2 H_2O$.

1. INTRODUCTION

The thermal decomposition of solid potassium tris(oxalato) ferrate(III) trihydrate has been extensively studied by several workers /1-8/ and various intermediate products and mechanisms have been suggested. If, in most cases, it was shown that, after dehydration at about $120^\circ C$, the anhydrous ferric complex is reduced into a ferrous species, the exact nature of the latter compound is still controverted. For instance, from X-ray diffraction (XRD) /2,4/ and infrared (IR) /3,6/ studies, it was clearly evidenced that the solid residue obtained after the second stage of decomposition at $300^\circ C$ is an heterogeneous mixture of anhydrous potassium oxalate and another component which was assumed to be $K_2Fe(ox)_2$, by analogy with similar cobalt(II) or manganese(II) /6/ complexes, but which was not directly identified because of the difficulty in its preparation.

Despite these pertinent observations, other interpretations of the nature of this residue have been reported from Mössbauer /5,8/ and thermal analysis measurements /7/, such as $K_6Fe_2(ox)_5$ /5,8/ and even (!) $K_3[Fe(ox)(CO_3)O]$ /7/.

In this work and for the first time, we succeeded in preparing pure $K_2Fe(ox)_2 \cdot 2 H_2O$ and $K_2Fe(ox)_2$ complexes and the thermal decomposition of these compounds has been compared to that of $K_3Fe(ox)_3 \cdot 3 H_2O$ up to $700^\circ C$, using thermogravimetric analysis (TGA) and differential scanning calorimetry (DSC) in order to determine the different decomposition stages. In addition, Mössbauer spectroscopy and XRD have been used to identify the exact nature of the solid residues after each stage.

2. EXPERIMENTAL

The complex $K_3Fe(ox)_3 \cdot 3 H_2O$ was prepared according to the literature /9/. Potassium di(oxalato) ferrate(II) dihydrate was prepared by dissolving 27 g of $K_2ox \cdot H_2O$ and 9 g of $Feox \cdot 2 H_2O$ in 150 mls of boiling water, under pure nitrogen atmosphere; after complete dissolution, the resulting red solution was evaporated up to a volume of 100 mls; on cooling down to room temperature gold-yellow crystals of $K_2Fe(ox)_2 \cdot 2 H_2O$ were precipitated from the solution, filtered off

and washed with water, alcohol and ether. As this compound is slightly oxidable, all the experiments have been carried out under pure nitrogen.

Details on the thermal analyses and Mössbauer measurements have been described previously /10/. The spectra were recorded at 296 and 80K and all isomer shifts (IS) mentioned in the text are quoted relative to natural iron at room temperature. XRD measurements have been performed on a Siemens 500 D diffractometer, using the copper K_{α} radiation.

3. RESULTS AND DISCUSSION

The TGA curve of $K_3Fe(ox)_3 \cdot 3H_2O$ (fig.1) shows four separate stages in the decomposition at about 113°C, 296°C, 470°C and 577°C. These transitions are in agreement with the presence of peaks AB, BC, CD and DE in the DSC curve (fig.1a), whereas the additional endothermic and reversible peak M at 400°C must be attributed to the melting of K_2Ox , since it doesn't involve a change in weight and corresponds fairly well to this characteristic phase transition in the pure potassium oxalate (fig.1b). This DSC result indicates clearly that K_2Ox is one of the components in residue C.

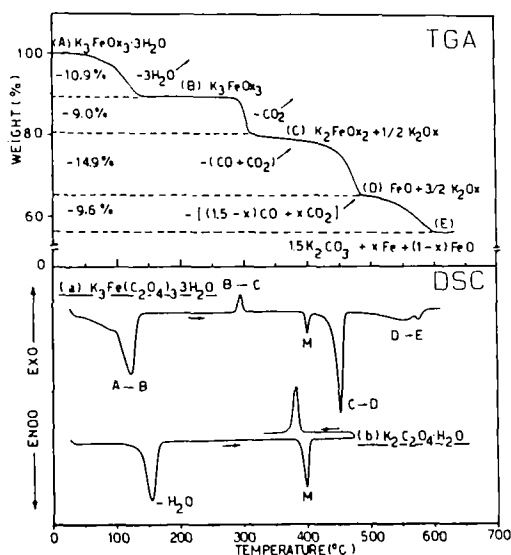


Figure 1. TGA and DSC curves of a) $K_3Fe(ox)_3 \cdot 3H_2O$ and b) $K_2(ox)H_2O$.

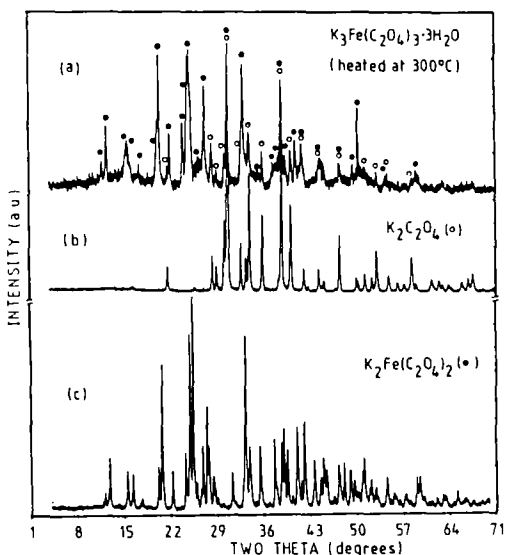


Figure 2. XRD analysis of a) $K_3Fe(ox)_3 \cdot 3H_2O$ heated at 300°C, b) K_2Ox and c) $K_2Fe(ox)_2$.

In agreement with earlier works /2,4/, the presence of K_2Ox (fig.2b) is also evidenced in the XRD pattern of residue C (fig.2a) which shows, in addition, another component identical to $K_2Fe(ox)_2$ (fig.2c). It is noteworthy that this orange-yellow complex is also obtained by dehydration of the dihydrated salt, $K_2Fe(ox)_2 \cdot 2H_2O$, at 200°C, according to the relative weight loss (w.l. = 10.3%) observed on the corresponding TGA curve (fig.3) and the calculated value for the loss of two water molecules (c.w.l. = 10.4%).

Similarly, the thermal analyses of $K_3Fe(ox)_3 \cdot 3H_2O$ confirm that the three molecules of water are evolved as a single and endothermic dehydration stage at 113°C (c.w.l. 10.99%). The Mössbauer spectrum of $K_3Fe(ox)_3$ (fig.4b) is characterized by a single broad line (1') whose parameters (IS = 0.31 ± 0.03 mm/s, line width, l.w. = 1.94 mm/s) are very similar to those of the trihydrate (fig.4a, line 1) (IS = 0.31 ± 0.03 mm/s, l.w. = 2.00 ± 0.05 mm/s).

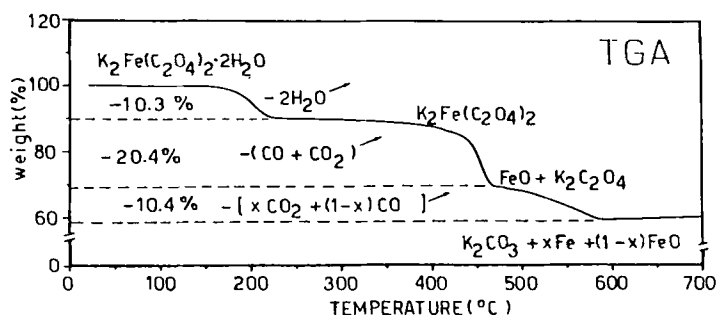


Figure 3. Thermogravimetric analysis of $K_2Fe(ox)_2 \cdot 2H_2O$ in nitrogen.

The second stage BC of decomposition at 296°C is due to an exothermic reduction of iron(III) to iron(II). The spectrum of the residue C obtained at 300°C (doublet 22', fig.4c), exhibits indeed an isomer shift ($IS = 1.15 \pm 0.03$ mm/s) and a temperature dependent quadrupole splitting ($QS = 3.89 \pm 0.03$ mm/s at 296K; $QS = 4.09 \pm 0.03$ mm/s at 80K) which are characteristic of high spin divalent iron. Since these parameters are identical to those of anhydrous $K_2Fe(ox)_2$ (fig.4f) obtained by dehydration of $K_2Fe(ox)_2 \cdot 2H_2O$ (figure 4e, doublet 33', $IS = 1.21 \pm 0.03$ mm/s, $QS = 2.56 \pm 0.03$ mm/s at 296K; $QS = 3.24 \pm 0.04$ mm/s at 80K), the former compound is definitely the iron-containing component of residue C. Moreover, as the observed w.l. (9.0%) is identical to the theoretical value (w.l. = 8.96%) for the loss of one CO_2 group per iron atom, the second stage of decomposition must be represented by the following mechanism :

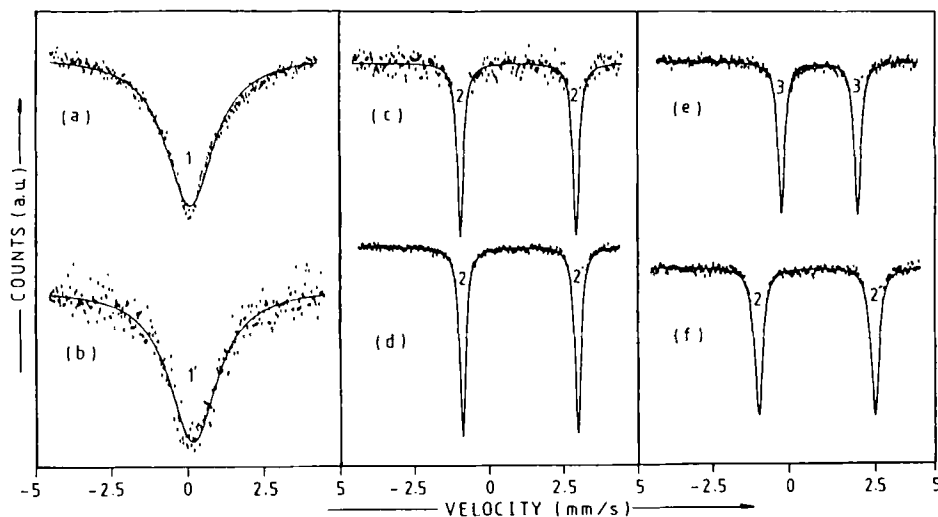
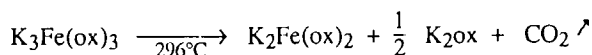


Figure 4. Mössbauer spectra at 296K of a) unheated and b) c) d) heated $K_3Fe(ox)_3 \cdot 3H_2O$ up to 150°C, 300°C, and 400°C, e) $K_2Fe(ox)_2 \cdot 2H_2O$ and f) $K_2Fe(ox)_2$.

It should be noticed that, in pure nitrogen, this ferrous species is stable up to 400°C, even after the melting (M) of K_2Ox , the second component of residue C (fig.4d).

Beyond this temperature, the further decomposition stages of both $K_3Fe(ox)_3 \cdot 3 H_2O$ and $K_2Fe(ox)_2 \cdot 2 H_2O$ are of course similar : after the decomposition of $K_2Fe(ox)_2$ at about 470°C into K_2Ox and FeO (which disproportionates partly into Fe_3O_4 , Fe and Fe_3C) (fig.5a), the potassium oxalate is decomposed at 577°C into potassium carbonate and carbon monoxide which causes a total reduction of Fe_3O_4 into FeO and metallic iron (fig.5b). This last mechanism is also supported by the observed weight loss (9,6%, fig.1) and the theoretical value (w.l. = 10,6%) calculated from the relative proportion of metallic iron found by Mössbauer spectroscopy in the final residue ($x = 0.63$).

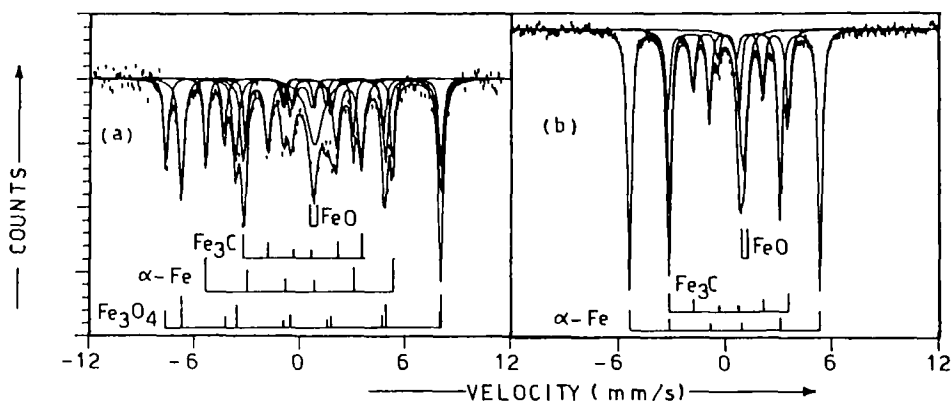


Figure 5. Mössbauer spectra at 296K of $K_3Fe(ox)_3 \cdot 3 H_2O$ heated at a) 500°C and b) 650°C under nitrogen.

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